

SUBROTA CHANDRA SARKER

Full Stack Web Developer

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PROFESSIONAL SUMMARY

Web Developer with 3+ years of experience building and scaling resilient web applications and distributed backend infrastructure. Specialized in high-throughput backend architecture, database optimization, and implementing complex business logic using a strong foundation in Data Structures and Algorithms (DSA). Experienced in modernizing legacy monolithic systems into highly scalable, modular platforms utilizing both JavaScript/TypeScript (MERN stack) and Python (Django), alongside deep expertise in relational database design (MySQL) and PHP environments. Strong hands-on ownership across system design, API engineering, deployment workflows, observability, and production reliability.

Core Competency

- API Design & Security • System Modernization • Efficiency Algorithmic Design (DSA) • Relational Database Optimization • MERN Stack (MongoDB, Express, React, Node.js) • Python & Django Ecosystem • Laravel & Enterprise PHP • Low-Level & Performance Programming (C/C++) • MySQL Database Architecture
- GitHub • Firebase • Bootstrap • Tailwind CSS

TECHNICAL STACK

Backend: PHP, Laravel, Node.js, Express.js, Django, REST API

Frontend: React.js, Next.js, Redux, TypeScript

Architecture: Clean Architecture, Hexagonal Architecture

Database: MySQL, MongoDB, PostgreSQL

KEY CONTRIBUTIONS & IMPACT

- Led the modernization of a legacy monolithic PHP/Laravel payment platform, architecting its decoupling into highly scalable, modular services using Clean Architecture and Hexagonal Architecture principles to improve long-term framework flexibility.
- Managed core backend systems across multiple payment gateways, utilizing a high-performance mix of Node.js (MERN stack) and Django
- Integrated and optimized local and international payment methods, including Visa, Mastercard, American Express, bKash, Nagad, Rocket, Upay, and automated bank deposit APIs.
- Leveraged core Data Structures and Algorithms (DSA) to design low-latency, memory-efficient state machines for transaction state management and custom idempotency mechanisms, entirely eliminating race conditions.
- Achieved a 60% reduction in platform latency while maintaining 99.99% uptime, utilizing low-level optimization principles (C/C++) for critical execution paths and deep database refactoring.
- Engineered and optimized MySQL database layers, drastically increasing throughput via strategic indexing, query execution tuning, dead-lock prevention, and eliminating expensive, non-scalable JOIN operations.
- Implemented centralized logging and observability using the ELK Stack (Elasticsearch, Logstash, Kibana) to monitor transaction health, API latencies, and critical error rates in real-time.
- Architected automated deployment pipelines using Docker containerization, Jenkins, and custom shell scripting to transition infrastructure smoothly from a legacy monolith to a containerized microservices setup.
- Led code reviews, architecture discussions, and technical mentoring for a 7-member engineering team, instilling strict standards for clean code, relational database integrity, and robust error handling.
- Co-authored additional enterprise microservices, including recurring subscription engines, automated notification systems, SendBill services, and backward-compatible REST APIs.

EDUCATION

Secondary School Certificate (S.S.C)

- Institute : Dadra High School
- Group : Science
- Result : GPA: 4.06 out of 5
- Board : Rajshahi
- Passing Year : 2018

Higher Secondary School Certificate (H.S.C)

- Institute : Jamalgonj Degree College
- Group : Science
- Result : GPA: 4.08 out of 5
- Board : Rajshahi
- Passing Year : 2020

Courses

- ❖ Programming Hero
- ❖ Phitron.io

ADDITIONAL INFORMATION

Available for full-time remote opportunities with overlap for EU and US time zones.